

MATTHEW MALONI

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EDUCATION

University of Florida, College of Liberal Arts and Sciences

August 2023 - May 2027

Bachelor of Science in Data Science

Gainesville, FL

- GPA: 3.63 | Dean's List
- **Relevant Coursework:** Applied Machine Learning Systems; Programming Fundamentals I and II; Programming with Data in R; Linear Regression; Probability; Linear Algebra for Data Science

WORK EXPERIENCE

Trustwell

April 2025 - August 2025

Data Science Intern

Salem, OR (Remote)

- Compiled regulatory datasets covering all 50 U.S. states plus North America and Europe, expanding training materials and significantly improving AI response quality for market compliance guidance.
- Designed custom prompt engineering strategies to optimize the AI agent's use of OpenAI API, enabling evidence-based recommendations for restaurant clients expanding nationally and internationally.
- Migrated web app components from AngularJS to Vue 3, enhancing functionality and performance.

LEADERSHIP & INVOLVEMENT

IEEE Signal Processing Society

September 2025 - Present

Research Board - Project Lead

Gainesville, FL

- **Plato's Cave - Scientific Paper Verification**
- Contributed to trust-propagating knowledge graph system for scientific paper verification, implementing scoring algorithms, edge confidence visualization, and trust-gating mechanisms; compiled evaluation dataset of 150+ papers and built testing pipelines validating node quality assessment.
- Co-authored research paper documenting evaluation methodology, ablation studies, and performance benchmarks demonstrating improvements over single-claim-based and LLM-only baselines.
- **Binaural Audio Systems (Project Lead)**
- Leading research group exploring spatial audio localization and source separation using dual-microphone systems for real-time speech separation and sound position detection in acoustic environments.

Nettles Lab, UF Scripps Institute

January 2026 - Present

Undergraduate Researcher

Jupiter, FL

- Contributing to NIH-funded drug discovery research targeting steroid hormone receptors, implementing self-supervised autoencoder pipeline to optimize predictive models across molecular datasets.

Beta Theta Pi Gamma Xi

August 2025 - Present

Internal Philanthropy Chair

Gainesville, FL

- Leading Florida Greek life's largest fundraising initiative, orchestrating campaign logistics, donor relations, and event coordination with 300+ participants to raise \$50,000 for the Alzheimer's Association.

PROJECTS

Pastry Art Wine Club - Full Stack Mobile Application

October 2025 - Present

- Built production mobile app (React Native/Expo) with OpenAI API integration for wine label analysis and personalized chatbot featuring web search capability; implemented fuzzy matching system.
- Designed user cellar management system with SQL backend handling authentication and data persistence; prepared application for App Store submission.

CNN Medical Image Classification Research

November 2024

- Architected and compared 4 CNN models using TensorFlow Keras for multi-label pathology detection, achieving 94.77% accuracy and 83% AUC on 14-disease classification with 78K images.

ChartScanAI Automated Trader

August 2025 - Present

- Integrated pretrained YOLOv8 model into trading workflow screening 500+ stocks; developed dynamic screening pipeline with yfinance data handling, backtesting, and paper-trading via Alpaca API.

SKILLS

Python, C++, R, SQL, Scikit-learn, PyTorch, TensorFlow, Git, AWS, Research Communication